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Promotion and caution in research article abstracts: The use of positive, negative and hedge words across disciplines and rankings

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Abstract: Abstract as a promotional genre has been an increasing interest in recent years, leading to an intriguing debate on the objectivity of scientific writing. The present study investigated the promotion and caution in research article abstracts through the use of positive, negative and hedge words across disciplines and rankings based on a large and principled dataset (more than 12.6 million words). The corpus was designed and built with full consideration of representativeness, structure, balance, and size in terms of discipline and ranking. The results showed that positive words were more frequently used than negative words in terms of both discipline and ranking, positive and negative words were more frequently used in hard sciences than those in soft sciences and high ranking journals than those in low ranking journals, and hedge words were more frequently used in high ranking journals and soft sciences. Further investigation also found a complexity of frequency patterns when two disciplines were broken down into specific categories. The more frequent use of positive words in abstracts to promote scientific research was discussed from the perspective of disciplinary knowledge construction in addition to the publication bias, the outcome reporting bias, and universal linguistic positivity bias.

Keywords: abstract, promotion, discipline, impact factors, linguistic devices

INTRODUCTION

Research article (RA) abstracts play two roles in academic publication. First, the RA abstract is an essential part of RA to summarize the idea of the whole article (Swales, 1990), making it 'an important genre to study' (Ebrahimi & Chan, 2015, p. 2). Second, the RA abstract also has the promotional role in 'persuading readers, and reviewers, to take the time to go further into the paper itself' (Jiang & Hyland, 2017, p. 1). According to Pho (2008), the schematic structures of RA abstracts were the main focus of most previous studies while just a few studies investigated the linguistic features of abstracts (p. 232). Specifically, the linguistic features of abstracts were studied either small in number (Pho, 2008, p. 232) or strictly linked to moves of abstracts (Pho, 2008, p. 232; Khany & Malmir, 2020, p. 4). Therefore, other linguistic features like the performance of positive and negative words in RA abstracts are still in need of extensive studies.

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Although there has been a long-standing debate on the distinction between soft and hard sciences (see Becher, 1994; Biglan, 1973; Hedges, 1987; Shapin, 2022; Storer, 1967), the concept provided a powerful tool for the classification of the knowledge production through comparative studies across disciplines. Many studies adopted Becher's (1989) taxonomy of disciplines as a standard to determine soft and hard sciences (e.g., Gao, 2016; Dang, 2018). For example, Gao (2016) investigated linking adverbials in research articles in soft and hard sciences and found that soft science articles contained more linking adverbials than hard sciences. By linking adverbials, Gao (2016) defined them as 'part of the cohesive devices', which both linguistically organize the structure of arguments and psychologically induce a certain interpretation of texts (p. 14). Dang's (2018) comparative studies on the vocabulary lists in academic speeches between soft and hard sciences found that the most frequent 3000 words were important for understanding both soft and hard sciences, but the value of these words in soft science was greater than in hard science. Although abstracts from various disciplines have been studied including applied linguistics (Dos Santos, 1996), medicine (Salager-Meyer, 1992), psychology (Hartley, 2003), and Business (Noguera, 2012), and so on, there is still a lack of attention on the comparison of studies from more disciplines in view of soft and hard sciences, especially on the positive and negative words across these disciplines.

Previous studies have found that human languages show a tendency towards using more positive words than negative words (Boucher & Osgood, 1969; Dodds et al., 2015). Further observations even demonstrated that the positivity bias also prevailed in scientific discourses (Bordignon et al., 2021; Cao et al., 2021; Vinkers et al., 2015). For example, Vinkers et al. (2015) compared 25 positive words and 25 negative words in scientific abstracts from medical journals between 1974 and 2014. Based on the same word lists of Vinkers et al. (2015), Cao et al. (2021) also revealed a positivity bias in both medical abstracts and articles in PubMed in terms of year-wise comparison between 1997 and 2019. By extending the number of positive and negative words in Vinkers et al. (2015), Bordignon et al. (2021) compared the positive words and negative words in medical abstracts of preprints between 2019 and 2020 in order to show the influence of the COVID-19 pandemic on the strategies of word choice. These studies have shown that the use of more positive words in scientific discourse seems to be definite. However, the tendency of positivity bias in other disciplines remains unclear.

According to Journal Citation Reports (JCRs) (https://clarivate.libguides.com/jcr), the Impact Factor (IF) of a journal is calculated by using the formular $IF = \frac{\text{the number of citations in the JCR year}}{\text{the total number of articles published in the two previous years}}$. Previous studies also found relationships between impact factors and certain aspects of articles, such as the correlation between impact factors and paper length (Xie et al., 2019), between impact factors and title length (Tempest, 2005), and between impact factors and word choice (Cao et al., 2021; Fraser & Martin, 2009). According to Cao et al. (2021), the correlation between positive words and journal impact factors was very weak, but the result seems contrary to Fraser and Martin (2009) that 'there was a significant increase in the use of highly biased words over the past 20 years in high impact journals but not in low impact journals' (p. 4). However, whether this relationship still holds or not in other disciplines requires further investigations.

Key points

- More positive words are used in abstracts than negative words in terms of both disciplines and ranking.
- More positive and negative words are used in hard sciences and high ranking journals.
- More hedge words are used in soft sciences and high ranking journals.
- A complex pattern of word frequencies in terms of fine disciplinary categories shows a great variance in view of caution, discipline, and ranking.

PROMOTION IN ACADEMIC WRITINGS

Although the tone of science writing should be objective (Barrass, 2002; Zobel, 2004), Pérez-Llantada (2003) argued that 'technical communication does not only involve transcribing data in a clear and objective way but also convincing the audience of the validity of certain claims and proposals' (p. 28). The myth of objectivity in science writing (Miller, 1969) has been studied under a wide range of rhetoric terms such as 'hype' (Hyland & Jiang, 2021; Millar et al., 2020), 'promotion' (Martín & Pérez, 2014; Wang & Yang, 2015), 'marketing' or 'selling' (Fraser & Martin, 2009; Moreno, 2021), and 'boosting' (Peacock, 2006), and so on. For example, boosting or boosters refer to 'pragmatic devices for emphasising certainty' by using linguistic expressions such as *clearly* and *obviously* (Peacock, 2006, p. 61). As Wang and Yang (2015) stated, the analysis of linguistic devices that encoded subjective attitude may further reveal the practice of promotion in RAs, 'whose socially recognized purpose is reporting research objectively' (p. 163). This line of study might be more enlightening when the disciplinary factor was considered due to the differences in knowledge construction across different disciplines (see Moyano, 2019).

Unlike boosters, hedge 'is an item of language which a speaker uses to explicitly qualify his/her lack of commitment to the truth of a proposition he/she utters' (Crompton, 1997, p. 282). In science writing, the use of hedge words by academic writers 'to downplay their statements and anticipate audience responses by adjusting the degree of certainty they give to their claims' seemed to be more cautious and modest (Hyland, 1994, p. 241). Therefore, the most important reason to study hedges lies in that the way academic writers using hedges in RA would provide 'insights into how science establishes its claims to knowledge, and how scientists carry out their work' (Hyland, 1998a, p. 7). Empirical studies on

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hedges had shown many details of hedges in scientific publications. For example, Hu and Cao's (2011) comparative study of hedges and boosters between the English-medium applied linguistics journals and the Chinese-medium counterparts revealed a cross-linguistic difference between empirical and non-empirical RA abstracts. Similarly, Yang (2013) also found cross-linguistic and cross-cultural differences across three sections of RAs, including Introduction, Method, and Results and Discussion. These studies observing the use of hedges against several contextual factors (Hyland, 2002) could uncover the frequency of hedges and/or boosters accordingly in a static way. That is to say, a possibly productive observation of the interaction between boosters and hedges was neglected. On the contrary, Bordignon et al. (2021) tried to explore the relationship between the use of hedge words and positive and negative words, thus providing a dynamic picture of word choices in the COVID-19 context. Although Bordignon et al. (2021) did not adopt a fine method specifically targeting this interactive aspect of wording behaviour, their idea and method could be applied further to other related studies because this line of study was just at its initial stage.

AIMS OF STUDY

To address the foregoing issues, the present study aims to find out the relationship between the use of positive, negative, and hedge words in RA abstracts across different disciplines and rankings (as measured by the journal's citations and Impact Factor). Embedding the results in the perspectives such as linguistic promotion and knowledge construction, it would answer the following research questions:

RQ1: *What are the patterns of frequency distribution in terms of positive, negative, and hedge words in RA abstracts?*

RQ2: *Is there any difference in the frequency of positive, negative, and hedge words in RA abstracts between high ranking and low ranking journals (as measured by the journal's citations and Impact Factor)? If so, what is the difference?*

RQ3: *Is there any difference in the use of positive, negative, and hedge words across disciplines? If so, what is the difference?*

The aim of the present study tries to explore both the internal relationship among word choices in terms of positive, negative, and hedge words and the external relationship among the choices of words and contextual factors such as discipline, ranking, and knowledge construction.

Following Bordignon et al. (2021), it should be noted that the aim of the present study is 'not to judge the tonality of the whole abstract' (p. 7), just to identify the positive, negative, and hedge words in the abstracts, thus differentiating it from the sentiment analysis of abstract (Wen & Lei, 2022).

METHODS

The workflow of the study contains three major steps: data collection, data processing, and data analysis. In the first step, the source of the dataset, criteria for data selection, and procedure of data collection are introduced. The second step illustrates the details of data cleaning and identification of positive, negative, and hedge words. In the last step, statistical methods used for the data analysis are presented. At the end of this section, further consideration of data collection was specified.

Data collection

The dataset of abstracts was collected from Web of Science (WoS) (www.webofscience.com/wos/woscc/basic-search), the oldest, most widely used, authoritative database of scientific publications and citations due to its selective, balanced, and complete coverage of world's leading journals (Birkle et al., 2020, p. 363). To be specific, Journal Citation Reports (https://jcr.clarivate.com/jcr/browse-categories) from Web of Science Core Collection categorized journals into 21 groups, and journal article abstracts were first confined to each of 19 groups. Two groups, 'Multidisciplinary' and 'Social Sciences, General', were excluded because journals from these two groups were highly overlapping with other specific groups. For example, categories within the 'Multidisciplinary' group, such as 'CHEMISTRY, MULTIDISCIPLINARY' and 'COMPUTER SCIENCE, INFORMATION SYSTEMS', were also included in the 'Chemistry' and 'Computer Science' groups, respectively. Furthermore, one or two categories that generally included the most journals from each group were selected, and especially two categories were selected when the group contained two related categories indicated by its title such as 'Economics & Business', 'History & Archaeology', and 'Plant & Animal Science', and so on. Categories containing the most journals were preferred due to the fact that journals from different categories within the same group were largely overlapped and that journals selected from different categories would not be duplicated across groups. Lastly, after sorting journals in either descending or ascending orders based on their impact factors or citations, 20 journals from each category were selected one after another in that order in terms of 10 high and 10 low journal impact factors (JIFs) for SCIE and SSCI journals or journal citation indicators (JCIs) for A&HCI journals. Because A&HCI journals did not have JIFs, JCIs were used for selecting A&HCI journals, which were 'designed to complement the JIF – the original and longstanding metric for journal evaluation' and the value of JCI was the mean Category Normalized Citation Impact (CNCI) for all articles and reviews published in a journal in the preceding 3 years (Szomszor, 2021). As a result, 540 journals from 27 categories became the source of abstracts. It should be noted that categories in JCRs were also labelled with index editions such as 'SCIE', 'SSCI', and 'AHCI', which was helpful to distinguish between hard and soft sciences.

In order to be comparable, the collection of abstracts should simultaneously apply the following three criteria:

1. all abstracts in the journals should be published in 2019;
2. the number of abstracts from each journal should be more than 20;

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3. all abstracts must come from the type of paper labelled as 'Research Articles'.

The data selection criteria were considered for the following reasons. First, confining abstracts published within a single year could avoid the influence of diachronic changes in linguistic choices (see Cao et al., 2021). Furthermore, the data from the year 2019, a period before the full-blown stage of the COVID-19 pandemic, would be more stable because the lexical strategies in abstracts were strongly influenced by the COVID-19 crisis (see Bordignon et al., 2021). Second, no less than 20 abstracts for each journal would ensure a considerable size of the dataset for comparative study because, in the pilot query, it was found that a number of journals, especially low-impact ones, published just a few articles. As a result, the total number of abstracts for each category would be at least 200 to meet the baseline of sentiment analysis (see Wen & Lei, 2022), although the present study was not to analyse the sentiments, attitudes, or emotions in abstracts, which normally required larger datasets (Serrano-Guerrero et al., 2015). Third, the abstracts exclusively from research articles were selected to guard against the potential influence of text types on vocabulary use (see Bal-Gezegin & Baş, 2020).

After determining the 20 high ranking and low ranking journals (as measured by the journal's citations and JIFs/JCIs) in 27 categories with index labels, the abstracts from each journal were manually collected from the Web of Science. First, the 'ISSN' or 'eISSN' was copied into the query box on the advanced search page by using the formula 'IS=ISSN/eISSN'. The results were refined by using 'Refine results' filter. To be specific, 'Publication Years' of '2019', 'Document Types' of 'Articles', and 'Languages' of 'English' were selected and refined, respectively. Simultaneously, 'Document Types' of 'Review Articles', 'Biography', 'Unspecified', 'Meeting', 'Case Report', 'Clinical Trial', 'News', 'Reference Material', 'Data Set', 'Expression Of Concern', and 'Report', but not limited to these types, were excluded. Second, after filtering the results to meet the data selection criteria, abstracts of each journal were exported to 'Excel' by selecting 'Record Content' as 'Author, Title, Source, Abstract'. Lastly, all abstracts were stored in one workbook with each category in one worksheet and each journal in one column.

Data cleaning

A small proportion of abstracts contained various copyright messages like '(c) 2019 Elsevier Inc. All rights reserved' and 'This is an open-access article under the CC BY licence. (http://creativecommons.org/licenses/by/4.0/)', and section labels in different forms and expressions such as 'Introduction' or 'Background', 'Methods' or 'Methodology', 'Results' and 'Discussion', and so on. These messages and words were manually deleted by using the 'Replace' dialogue box after a close examination of each 'Find'.

Positive, negative, and hedge words

Based on Vinkers et al. (2015) and Bordignon et al. (2021), the lists of positive, negative, and hedge words were adapted for the purpose of the present study. Inspired by Bordignon et al. (2021), the lists of positive, negative, and hedge words were modified in order to be balanced, comprehensive, and representative. The positive word list was extended to 60 plus additional 12 expressions of *significant* in positive form (Table 1). The extension of this list was based on the exhaustive examination of the New General Service List (NGSL; Browne et al., 2013) and the Academic Word List (AWL; Coxhead, 2000) and then was broadened through the online dictionaries of synonyms such as Cambridge (https://dictionary.cambridge.org/thesaurus/) and Merriam-Webster (www.merriam-webster.com/thesaurus). The NGSL had wide coverage in four major corpora (Brezina & Gablasova, 2015), and the AWL was developed from a balanced and representative Academic Corpus containing four disciplines: Arts, Commerce, Law, and Science. That is to say, the words from this list would not be biased for the present study.

In a similar vein, 60 negative words plus additional 12 expressions of *significant* in negative form were listed based on the same procedures, but one additional point kept in mind was that some negative words were added to the list as the counterparts of the positive words, that is, the antonyms (Table 2).

The listing of hedge words was conducted with reference to Abdollahzadeh (2019). As a result, 160 hedge words were included in the list based on the original 111 terms of hedges in Bordignon et al. (2021) (Table 3).

It was worth noting that all positive, negative, and hedge words in the lists were collaboratively determined with the aid of another scholar in English literature studies from the lists of candidate words out of the preliminary examination of the previous studies and online searching. Another notice was that the expressions of *significant* in both positive and negative lists excluded the adverb forms in Bordignon et al. (2021), such as 'are/is/were/was (not) significantly improved' in order to be consistent in terms of part of speech. Next, although three lists of positive, negative, and hedge words were enlarged, studies from sentiment analysis demonstrated that the total numbers both positive and

TABLE 1 Positive word list.

amazing, assuring, astonishing, astounding, bewildering, breathtaking, bright, creative, effective, efficient, encouraging, enormous, excellent, extraordinary, fantastic, favourable/favourable, groundbreaking, hopeful, impressive, incredible, ingenious, innovative, inspiring, inventive, marvellous/marvellous, miraculous, novel, outstanding, phenomenal, prodigious, prominent, promising, reassuring, remarkable, robust, satisfying, sensational, spectacular, staggering, startling, striking, stunning, stupefying, stupendous, substantial, successful, supportive, unique, unparalleled, unprecedented, wonderful, wondrous, advanced, beneficial, brilliant, constructive, fascinating, helpful, powerful, sufficient, is significant, are significant, was significant, were significant, is highly significant, are highly significant, was highly significant, were highly significant, is statistically significant, are statistically significant, was statistically significant, were statistically significant.

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TABLE 2 Negative word list.

detrimental, disappointing, disconcerting, discouraging, disheartening, disturbing, frustrating, futile, hopeless, impossible, inadequate, ineffective, insignificant, insufficient, irrelevant, mediocre, pessimistic, substandard, unacceptable, unpromising, unsatisfactory, unsatisfying, useless, weak, worrisome, inefficient, ambiguous, biased, defective, deficient, doubtful, faulty, flawed, impracticable, imperfect, imprecise, inaccurate, incoherent, inconsistent, indefensible, indistinct, negligible, obscure, powerless, uncertain, unconvincing, uncreative, undesirable, unfavourable/unfavourable, unfounded, unhelpful, unimportant, unimpressive, unjustified, unpersuasive, unpleasant, unreliable, unsuccessful, untenable, vague, is not significant, are not significant, was not significant, were not significant, is not highly significant, are not highly significant, was not highly significant, were not highly significant, is not statistically significant, are not statistically significant, was not statistically significant, were not statistically significant.

TABLE 3 Hedge word list.

a bit, about, actually, allege, almost, always, and so forth, apparently, appear, approximately, around, assume, at least, basically, be sure, believe, bunch, can, certainly, clear, conceivably, consider, consistent with, could, couple, definite, diagnostic, do not know, doubt, effectively, estimate, evidently, fairly, few, find, frequently, generally, guess, hopefully, improbable, in general, in my mind, in my opinion, in my understanding, in my view, inconclusive, indicate, kind of, largely, like, likely, little, look like, mainly, many, may, maybe, might, more or less, most, mostly, much, must, my impression, my thinking is, my understanding is, necessarily, occasionally, often, overall, partially, perhaps, possibility, practically, presumable, pretty, probability, quite, rarely, rather, read, really, roughly, say, seem, seldom, several, should, so far, some, somebody, somehow, someone, something, sometimes, somewhat, somewhere, sort of, speculate, suggest, suppose, surely, tend, think, understand, unlikely, unsure, usually, virtually, will, would, apparent, arguably, argue, attempt, cannot, certain, claim, contend, deduce, demonstrate, discern, essential, essentially, expect, expected, evident, feel, hardly, imagine, imply, infer, in line with, in theory, normally, note, observe, partly, possible, possibly, potentially, presumably, probably, propose, prove, relative, relatively, report, said, saw, see, seemingly, shall, show, slight, slightly, sought, surmise, theoretically, typically.

negative should be the same to prevent the possible inaccuracy of results due to unequal numbers of positive and negative lexicons (Ribeiro et al., 2016, p. 21; Taboada et al., 2011, p. 297). In other words, the lists in the present studies both added more words as in Bordignon et al. (2021) and kept a balance in their numbers as in Vinkers et al. (2015). Lastly, the lists, to our knowledge, were the most representative, balanced, and comprehensive for the purpose of the present study.

Statistical methods

All the raw data were normalized before conducting statistical analysis. In order to achieve the comparability between frequencies of different words in the lists, the normalized frequencies per million words of positive, negative, and hedge words in abstracts in terms of hard/soft, edition, and ranking were computed by using the formula $\mathrm{NF} = \frac{\text{the number of(P, N, or H) words in the abstract}}{\text{the total number of words in the abstract}} \times 100{,}0000$. MANOVA was used to test the significance of the overall results. Independent samples *t*-tests were performed to examine whether there were differences between disciplines (categories) and rankings (high and low-impact journals) in terms of normalized frequencies of the positive, negative, and hedge words in abstracts, respectively. Correspondence analysis was performed to explore the relationship between high or low ranking categories and the use of positive, negative, and hedge words.

Further consideration of data collection

Further consideration should be made to some details of data collection to complement the above-mentioned three steps. First, JIFs of 2019 were used to classify the SCIE and SSCI journals into high and low ranking groups while JCIs of 2020 were used for reference to classify the A&HCI journals. Both JIFs and JCIs could sort the journals in either ascending or descending orders on the WoS platform. Furthermore, all journals with high impact factors from SCIE belonged to Q1 in terms of JIF Quantiles, and those with low impact factors Q4. But high and low ranking groups from SSCI also included a few Q2 and Q3 journals in view of the criteria of abstract collection. When applying the criteria to the selection of journals within a particular category from SSCI, there was a shortage of journals within the scope of Q1 and Q4. Therefore, some journals from Q2 were selected based on the descending order of impact factors and added to journals selected from Q1, which went into high ranking group in the dataset; and some journals from Q3 were selected based on the ascending order of impact factors and added to journals selected from Q4, which went into low ranking group in the dataset. Similarly, considering the JCIs of A&HCI journals, most journals with high and low JCIs were located within the first quantile and the fourth quantile of that discipline based on the manual calculation of JCI quantiles, respectively. Some A&HCI journals were also selected from the second and third quantiles of that discipline due to the strict application of the criteria of abstract collection. Because these additional journals were small in number and close to Q1 and Q4 journals in terms of their JIFs or JCIs, they could be directly counted as high ranking journals and low ranking journals. In total, the corpus (more than 12.6 million words) consisted of 64,748 abstracts from 27 categories with high and low citations. Eventually, the corpus of the present study was principled with full consideration of representativeness, structure, balance, and size in terms of disciplines and ranking. Table 4 shows the descriptive statistics of the corpus, and detailed information about the corpus is presented in Appendix S1.

Second, it should be pointed out that the abstracts might be very slightly different if replication studies were conducted because the data collection of the present study lasted several

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TABLE 4 Descriptive statistics of the corpus in the present study.

Group	Category	Number of journals	Ranking	Number of abstracts	Number of words	Mean length of abstracts
Agricultural Sciences	AGRONOMY - SCIE	10	High	3117	763,354	244.90
		10	Low	901	185,496	205.88
Arts & Humanities, Interdisciplinary	HUMANITIES, MULTIDISCIPLINARY - AHCI	10	High	300	49,249	164.16
		10	Low	252	35,356	140.30
Biology & Biochemistry	BIOCHEMISTRY & MOLECULAR BIOLOGY - SCIE	10	High	606	120,610	199.03
		10	Low	536	91,900	171.46
Chemistry	CHEMISTRY, ANALYTICAL - SCIE	10	High	5821	1,134,045	194.82
		10	Low	1307	241,328	184.64
Clinical Medicine	ONCOLOGY - SCIE	10	High	480	123,804	257.93
		10	Low	1569	352,661	224.77
	SURGERY - SCIE	10	High	907	224,278	247.27
		10	Low	696	149,444	214.72
Computer Science	COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE - SCIE	10	High	2053	387,445	188.72
		10	Low	884	156,009	176.48
Economics & Business	BUSINESS - SSCI	10	High	584	95,235	163.07
		10	Low	406	66,113	162.84
	ECONOMICS - SSCI	10	High	913	134,165	146.95
		10	Low	362	45,702	126.25
Engineering	ENGINEERING, ELECTRICAL & ELECTRONIC - SCIE	10	High	4969	821,775	165.38
		10	Low	957	145,138	151.66
Environment/Ecology	PUBLIC, ENVIRONMENTAL & OCCUPATIONAL HEALTH - SCIE	10	High	443	103,558	233.77
		10	Low	668	149,812	224.27
	GEOGRAPHY - SSCI	10	High	1150	217,265	188.93
		10	Low	304	53,130	174.77
Geosciences	GEOSCIENCES, MULTIDISCIPLINARY - SCIE	10	High	1792	450,164	251.21
		10	Low	1212	229,612	189.45
History & Archaeology	HISTORY - AHCI	10	High	410	62,495	152.43
		10	Low	339	45,682	134.76

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	ARCHAEOLOGY - AHCI	10	High	720	133,560	185.50
		10	Low	287	44,531	155.16
Literature & Language	LITERATURE - AHCI	10	High	301	46,905	155.83
		10	Low	316	50,719	160.50
	LINGUISTICS - SSCI	10	High	453	80,574	177.87
		10	Low	347	58,144	167.56
Materials Science	MATERIALS SCIENCE, MULTIDISCIPLINARY - SCIE	10	High	2843	500,930	176.20
		10	Low	1007	159,319	158.21
Mathematics	MATHEMATICS - SCIE	10	High	1557	170,218	109.32
		10	Low	821	73,709	89.78
Philosophy & Religion	RELIGION - AHCI	10	High	942	154,510	164.02
		10	Low	294	40,098	136.39
	PHILOSOPHY - AHCI	10	High	657	94,228	143.42
		10	Low	254	39,228	154.44
Physics	ASTRONOMY & ASTROPHYSICS - SCIE	10	High	10,831	2,459,235	227.06
		10	Low	600	94,823	158.04
Plant & Animal Science	PLANT SCIENCES - SCIE	10	High	3059	758,684	248.02
		10	Low	563	102,320	181.74
	ZOOLOGY - SCIE	10	High	1062	250,001	235.41
		10	Low	382	65,096	170.41
Psychiatry/Psychology	PSYCHIATRY - SSCI	10	High	621	139,177	224.12
		10	Low	364	64,939	178.40
	PSYCHOLOGY, MULTIDISCIPLINARY - SSCI	10	High	1016	171,154	168.46
		10	Low	407	59,052	145.09
Visual & Performing Arts	COMMUNICATION - SSCI	10	High	721	110,201	152.84
		10	Low	385	61,008	158.46

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months and the completion of data collection was in December 2021. The main factor causing such a slight difference came from the search results which might include different 'online first' articles at different time points.

Third, due to the fact that a lot of journals from the same discipline might be indexed by different citation editions, such as 'AGRICULTURAL AND FOREST METEOROLOGY (ISSN 0168-1923)' was simultaneously indexed in order of relevance in 'AGRONOMY – SCIE', 'METEOROLOGY & ATMOSPHERIC SCIENCES – SCIE', and 'FORESTRY – SCIE', it was the first category that served as the decisive factor to determine the selection of the journal. In this case, the journal, 'AGRICULTURAL AND FOREST METEOROLOGY (ISSN 0168-1923)', was selected. This standard would also avoid overlapping in journal selection.

Fourth, four words in the positive word list and one word in the negative word list were treated specially. Two positive words, 'novel' and 'advanced' were tagged and counted in their adjective forms. Two other positive words were counted in their different spelling forms, that is, 'favourable/favourable' and 'marvellous/marvellous'. One negative word, 'obscure', was also tagged and counted in its adjective form.

Last, the data processing was conducted by using programming languages, such as Python and R. The homemade Python scripts were used to process texts including POS tagging and word frequency counting. Statistical analyses were performed by using R and SPSS, such as MANOVA by SPSS 22.0, *t*-tests and correspondence analysis by R.

RESULTS

Patterns of frequency distribution

Table 5 shows the normalized frequencies per million words of positive, negative, and hedge words in abstracts in terms of hard/soft, edition, and ranking. The pattern is clear that positive words appear more in the abstracts than negative words along each of three factors. But abstracts from high ranking journals and hard sciences contain more positive words than their counterparts, and the pattern is also true to negative words along the same levels. Similarly, the breakdown of soft sciences into SSCI and AHCI indicates that the former has higher frequencies of both positive and negative words than the latter. It is interesting that though abstracts from hard sciences contain more negative words than those from soft sciences, the breakdown of soft sciences shows that negative words in SSCI journal abstracts are the most among the three editions. As to hedge words, abstracts from SSCI and AHCI journals in soft sciences use more than SCIE in hard sciences, with the use of hedge words in AHCI journals being the most, while hedge words in high ranking journals are more than those in low ranking journals. In sum, the tendency is that the more positive words are used, the more hedge words are used.

A further breakdown of the word frequencies in terms of disciplines was presented in Fig. 1. The frequency pattern of positive, negative, and hedge words in abstracts across 27 disciplines indicates that 'Philosophy' (38,563.99) receives the highest frequency in using hedge words followed by 'Economics' (26,862.91) and 'Astronomy' (23,815.04), while 'Chemistry' (12,093.15) has the lowest frequency of hedge words in abstracts accompanied by 'Oncology' (12,845.94) and 'Surgery' (12,999.90). In terms of positive words, the highest frequency goes to 'Materials' (6,826.89), which is far more than 'Computer' (4,680.79) and 'Chemistry' (4,589.77), in contrast to the lowest frequencies in 'Environment' (1,219.84), 'Archeology' (1,370.55), and 'Zoology' (1,453.64). The most use of negative words belongs to 'Mathematics' (872.34), 'Philosophy' (866.02), and 'Computer' (830.77), with the frequency of negative words from 'Archeology' (147.75), 'Chemistry' (193.55), and 'Agronomy' (215.32) being the least. In sum, Fig. 1 shows a greater variation in the use of hedge and positive words across disciplines than that of negative words (Norm_P = Normalized frequency of positive words, Norm_N = Normalized frequency of negative words, Norm_H = Normalized frequency of hedge words).

The use of positive, negative, and hedge words showed a great variance in terms of journal ranking both within and across editions of journals (Fig. 2). To be specific, in terms of high ranking journals, abstracts from 'SCIE' use more positive (3383.78) and negative (482.85) words than those from 'SSCI' (2299.90/462.42) and 'AHCI' (1650.12/392.27), but they use fewer hedge words (18,809.70) than those from 'SSCI' (21,804.73) and 'AHCI' (25,218.09). In other words, 'SCIE' journal abstracts have the least frequency of hedge words and 'AHCI' journal abstracts the most. In terms of low ranking journals, the normalized frequencies of positive, negative, and

TABLE 5 Frequency of positive, negative, and hedge words.

Factors	Levels	Positive words	Negative words	Hedge words	Ratio between P, N, and H
Hard/soft	Hard	3047.83	437.59	17,762.55	6.97:1:40.59
	Soft	1955.49	428.10	22,275.64	4.57:1:52.03
Edition	SCIE	3047.83	437.59	17,762.55	6.97:1:40.59
	SSCI	2150.77	456.57	21,555.60	4.71:1:47.21
	AHCI	1646.22	383.01	23,416.05	4.3:1:61.14
Ranking	High	3141.91	474.30	19,589.52	6.62:1:41.3
	Low	1899.13	321.97	15,987.31	5.9:1:49.65

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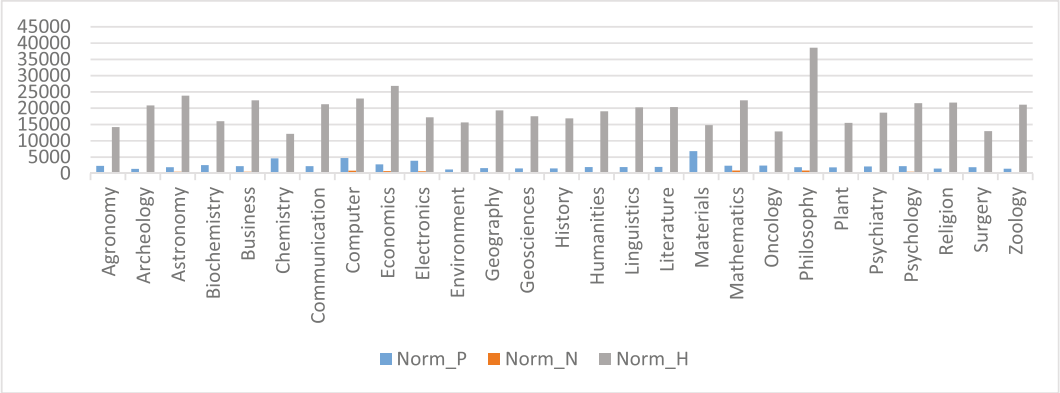


FIGURE 1 Disciplinary distribution of positive, negative, and hedge words.

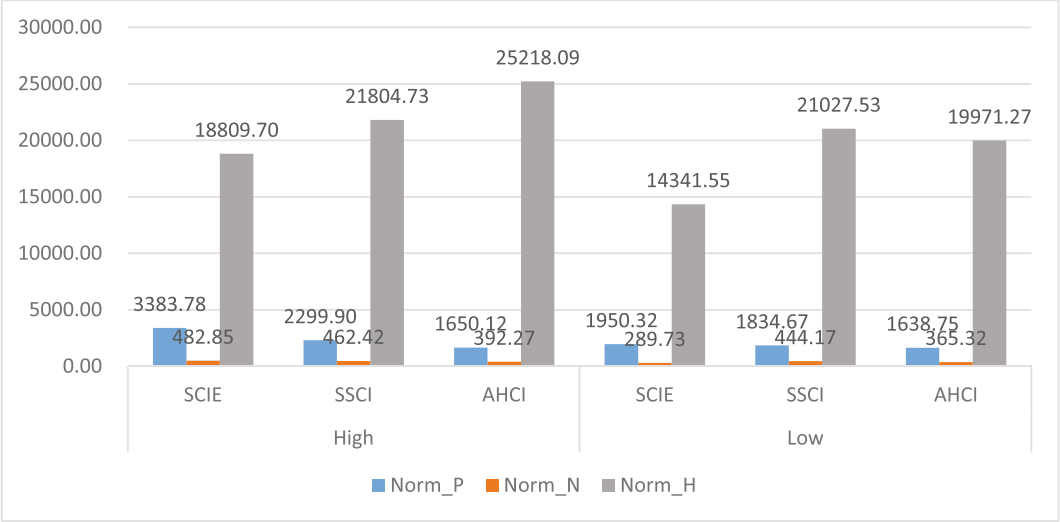


FIGURE 2 Distribution of positive, negative, and hedge words in terms of ranking.

hedge words across three editions show a different pattern from those in terms of high ranking journals. Although abstracts from 'SCIE' journals receive the most positive words (1950.32) followed by those from 'SSCI' journals (1834.67) and 'AHCI' journals (1638.75), abstracts from 'SSCI' journals write the most negative (444.17) and hedge (21,027.53) words. The most striking difference lies in the fact that abstracts from 'SCIE' journals in the low ranking section have the least frequencies of both negative (289.73) and hedge words (14,341.55). In terms of cross-sectional observation of both high and low ranking journals, similar patterns of distribution are retained across editions. In other words, the frequencies of positive, negative, and hedge words appear in descending order across 'SCIE', 'SSCI', and 'AHCI' journal abstracts respectively. That is to say, as shown in Fig. 2, the positive, negative, and hedge words are used more in 'SCIE', 'SSCI', and 'AHCI' journal abstracts in high ranking sections than those in low ranking sections with negative words being the least among three lists of words.

To further answer RQ1, correspondence analysis was performed with R programming language to show a detailed relationship between the use of positive, negative, and hedge words along 27 categories with high and low citations. CA function of FactoMineR package analysed the data and fviz_ca_biplot function of factoextra package visualized the results. The application of correspondence analysis depends on the significant dependency between the rows and columns, which is achieved by the chi-square test for the association between the row and column variables (Litkowski, 2020). In this study, the association is highly significant (chi-square = 41,974, $p < 0.00$).

Correspondence analysis could project row and column variables onto a two-dimensional plot (Fig. 3). Two dimensions could account for 100% of the information in the data, being regarded as effective. This is further supported by the significant test of the result of correspondence analysis by evaluating the degree of significant dependency between the rows and columns (chi-square = 41,973.79, $p < 0.00$).

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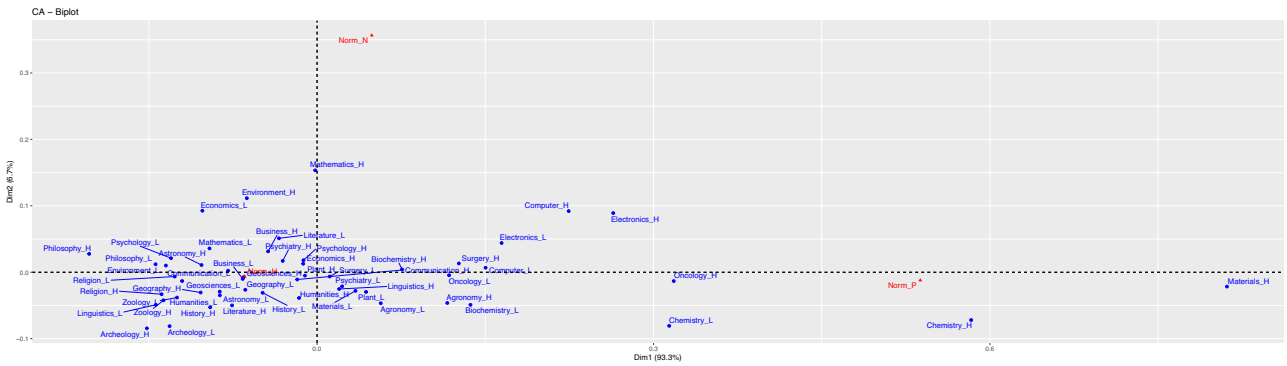


FIGURE 3 Correspondence analysis of category and impact.

Correspondence analysis shows a contrasting distribution of word choices and categories with high and low citations. To be specific, positive words seem to be associated with high ranking journals especially those in 'Chemistry', 'Materials', and 'Oncology', and negative words seem to be close to 'Computer', 'Electronics', 'Mathematics', and 'Environment' in high ranking section as well. Although the use of hedge words appears to be strongly connected with low ranking journals and soft sciences concerning 'Communication' with low impact factors, 'Business' with low impact factors, 'Religion' with low impact factors, and 'Geography' with low impact factors, and so on, some categories, such as 'Geosciences' with high impact factors, 'Plant' with high impact factors, and 'Astronomy' with high impact factors, also contains more hedge words in their abstracts.

A two-way MANOVA test with disciplines (Hard and Soft sciences) and ranking (High and Low citations) as the independent variables and the use of positive, negative, and hedging words as the dependent variables was performed to answer the RQ2 and RQ3, that is, whether there are differences among the use of positive, negative, and hedge words in terms of discipline and ranking, respectively, and if any, what the patterns of the difference are. Significant multivariate main effects were found for discipline with a small effect size ($F(364,742) = 504.54$, $p < 0.00$, $\eta_p^2 = 0.02$) and for ranking with a small effect size ($F(364,742) = 301.13$, $p < 0.00$, $\eta_p^2 = 0.01$), and also for the interaction between discipline and ranking with a small effect size ($F(364,742) = 60.36$, $p < 0.00$, $\eta_p^2 = 0.003$). Therefore, the null hypothesis was rejected and there were significant differences in the use of positive, negative, and hedge words in terms of discipline and ranking.

A further investigation of different word lists across disciplines and rankings reveals more details of the use of positive, negative, and hedge words, respectively. Independence sample *t*-test was performed to triangulate the two-way MANOVA test. Generally, the use of three different word lists shows cross-disciplinary and cross-ranking differences but all with small effect sizes, which was calculated by an online calculator using *t* and *df* as input values (www.99cankao.com/statistics/effect-of-size-calculator.php). First, the use of positive words between 'Hard' and 'Soft' sciences differs significantly with a small effect size [$t = 24.84$, $df = 23,499.00$, $p < 0.00$, 95% CI = (1006.13, 1178.55), $r = 0.16$]. The cross-ranking difference between the use of the positive words in 'High' and 'Low' impact journals is also significant with a small effect size [$t = 31.03$, $df = 35,718.00$, $p < 0.00$, 95% CI = (1164.28, 1321.29), $r = 0.16$]. Second, the use of negative words is also significantly different with a small effect size between high and low ranking journals ('High' vs. 'Low') [$t = 9.15$, $df = 33,513.00$, $p < 0.00$, 95% CI = (119.71, 184.95), $r = 0.05$], but non-significantly different between disciplines ('Hard' vs. 'Low') [$t = 0.49$, $df = 20,687.00$, $p = 0.62$, 95% CI = (−28.44, 47.42), $r = 0.00$]. Third, the use of hedge words has a similar pattern to that of positive words in term of both cross-disciplines ('Hard' vs. 'Low') [$t = -29.80$, $df = 17,642.00$, $p < 0.00$, 95% CI = (−4809.91, −4216.27), $r = 0.22$] and cross-ranks ('High' vs. 'Low') [$t = 29.23$, $df = 28,394.00$, $p < 0.00$, 95% CI = (3360.68, 3843.73), $r = 0.17$].

However, there are also non-significant differences across disciplines and rankings except for the use of hedge words when the disciplinary factor was further divided into three different editions, that is 'SCIE', 'SSCI', and 'AHCI'. The cross-disciplinary differences between positive and negative words seem to be larger than their cross-ranking differences. To be specific, the only non-significant difference in terms of positive words is that between abstracts from high and low ranking 'AHCI' journals [$p = 0.92$]. Contrary to the use of positive words, the use of negative words shows more non-significant results in terms of three editions. In this case, the disciplinary differences seem to be smaller than the cross-ranking differences. It is non-significant between 'SCIE' and 'SSCI' [$p = 0.43$], between 'SCIE' and 'AHCI' [$p = 0.05$], between 'SSCI' with high and low citations [$p = 0.71$], and between 'AHCI' with high and low citations [$p = 0.62$]. Moreover, the differences between 'SSCI' and 'AHCI' in terms of positive words [$p = 0.03$] and between 'SSCI' with high and low citations in terms of hedge words [$p = 0.03$] are narrowly significant (Table 6).

DISCUSSION

The present study investigated the use of positive, negative, and hedge words in research article abstracts. Linguistic devices as a

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promotional strategy might be, for the first time, examined against fine contextual factors such as discipline and ranking based on a large and principled dataset (over 12.6 million words). The main findings of the present study showed a complex relationship between these contextual factors and the choice of words, which were summarized in the following tendencies. First, positive words were used more frequently than negative words across disciplines and rankings. Second, positive words in high ranking journals were used more frequently than those in low ranking journals, and negative words in high ranking journals were also used more frequently than those in low ranking journals. Third, the use of positive words in detail showed bigger cross-disciplinary differences than cross-ranking ones, while the use of negative words in detail showed some non-significant differences in terms of both discipline and ranking. Fourth, hedge words were used more frequently in soft sciences than those in hard sciences, and they were also more frequently used in high ranking journals than those in low ranking journals. The use of hedge words in detail showed considerable differences in terms of both disciplines and ranking. Fifth, the use of positive, negative, and hedge words showed different relations to specific high or low ranking categories. The following detailed discussion of results will move from an overall interpretation of the vocabulary tendency to the specific interpretations of the patterns of three word lists across disciplines and ranking, which will lead to the implications and limitations of the present study.

Overall discussion of results

The propensity for positive words rather than negative words across disciplines and rankings and the relationship among positive, negative, and hedge words could be explained from the following perspectives. First, the current study regards the use of more positive words as an extension of the Pollyanna hypothesis into academic discourse, a view believing that 'there is a universal human tendency to use evaluatively positive (E+) words more frequently' (Boucher & Osgood, 1969, p. 1). The hypothesis was supported by a study with a large dataset, which claimed a

TABLE 6 Differences between the use of positive, negative, and hedge words.

Words	*t*-test	*t*	*df*	*p*	95% CI	*r*
Positive	Hard∼Soft	24.84	23,499.00	0.00	1006.13–1178.55	0.16
	SCIE∼SSCI	16.26	11,547.00	0.00	788.93–1005.20	0.15
	SCIE∼AHCI	23.28	6863.40	0.00	1283.59–1519.64	0.27
	SSCI∼AHCI	6.72	11,761.00	0.00	357.32–651.79	0.06
	High∼Low	31.03	35,718.00	0.00	1164.28–1321.29	0.16
	SCIE_H∼SCIE_L	31.13	26,046.00	0.00	1343.19–1523.73	0.18
	SSCI_H∼SSCI_L	4.39	5266.40	0.00	257.69–672.77	0.06
	AHCI_H∼AHCI_L	0.10	3577.70	0.92	−217.94–240.68	0.00
Negative	Hard∼Soft	0.49	20,687.00	0.62	−28.44–47.42	0.00
	SCIE∼SSCI	−0.78	10,670.00	0.43	−66.61–28.66	0.01
	SCIE∼AHCI	1.97	6295.00	0.05	0.25–108.90	0.02
	SSCI∼AHCI	2.12	11,423.00	0.03	5.64–141.46	0.02
	High∼Low	9.15	33,513.00	0.00	119.71–184.95	0.05
	SCIE_H∼SCIE_L	10.38	24,884.00	0.00	156.66–229.58	0.07
	SSCI_H∼SSCI_L	0.38	5052.50	0.71	−76.71–113.21	0.01
	AHCI_H∼AHCI_L	0.50	3745.20	0.62	−79.17–133.06	0.01
Hedge	Hard∼Soft	−29.80	17,642.00	0.00	−4809.91 to −4216.27	0.22
	SCIE∼SSCI	−21.78	10,043.00	0.00	−4134.38 to −3451.71	0.21
	SCIE∼AHCI	−21.94	5602.60	0.00	−6158.56 to −5148.43	0.28
	SSCI∼AHCI	−6.19	9277.10	0.00	−2449.26 to −1271.64	0.06
	High∼Low	29.23	28,394.00	0.00	3360.68 to 3843.73	0.17
	SCIE_H∼SCIE_L	34.10	20,503.00	0.00	4211.28 to 4725.02	0.23
	SSCI_H∼SSCI_L	2.18	4914.40	0.03	78.67 to 1475.75	0.03
	AHCI_H∼AHCI_L	10.50	4031.80	0.00	4267.31 to 6226.34	0.16

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universal positive bias with caution because this system-level property did not assert all natural texts present positivity bias (Dodds et al., 2015). The positivity bias was first found in academic discourse by Vinkers et al. (2015) and later verified by Cao et al. (2021) and Bordignon et al. (2021). The present study also evidenced that positivity bias exists in academic discourse.

Second, the inclination to use more positive words in abstracts might be caused by the so-called publication bias and outcome report bias in both hard and soft sciences (Dwan et al., 2013; Franco et al., 2014). Many studies have proved that studies reporting positive or significant results are more likely to be published (Begg & Berlin, 1988; Dwan et al., 2013; Easterbrook et al., 1991; Gerber & Malhotra, 2008). Other studies found that positive or significant outcomes are more likely to be fully reported (Dwan et al., 2013; Pigott et al., 2013; Williamson et al., 2005). The higher odds of positive results in the abstracts indicate a higher probability of linguistical encoding with positive words.

Third, the use of more positive words in abstracts might be related to the rhetorical strategy by using linguistic devices to promote 'the bright side' (Cao et al., 2021, p. 4) to attract potential audiences at the time of either reviewing or published status. This goes further to over-promotion (Bordignon et al., 2021), overstatement (Shinohara et al., 2017), or even hype (Millar et al., 2019). The motive for hype in academic discourse is underlaid by Hyland and Jiang's (2021) observation that 'academics are driven by pressures to gain visibility through their publications and citations as much as by the desire to get their research accepted' (p. 201), which helps to highlight 'the distinctiveness and value of authors' research in an increasingly competitive environment' (p. 201). This latter point of view leads the exploration of linguistic hype to a broader context, in which 'the complex array of social forces' contributes to the phenomenon of hype (Caulfield & Condit, 2012, p. 209). This entrenched 'hype pipeline' includes 'the pressure to publish, the increasingly intense commercialization agenda, the messaging emanating from research institutions, the news media and, even, the public itself' (Caulfield & Condit, 2012, p. 209).

Fourth, the correspondence analysis showing a more complicated picture of positive, negative, and hedge words in terms of disciplinary and ranking factors engenders a more critical view towards the fine details of interaction among three factors. Triangular distribution of positive, negative, and hedge words indicating a difference in usage also reveals a tension that the use of positive and hedge words together has an equal opportunity to the use of negative and hedge words together in general, which seems to be contrary to the previous study (Bordignon et al., 2021). The tension balances two possible patterns of word distribution: hedge words have a bigger chance to modify positive words and they also tend to modify negative words more frequently. Hedge words as one of the linguistic devices to realize mitigation serves to modify certain unwelcome effects of speech acts on the hearer (Fraser, 1980). It is definitely true of the potentially unwelcome effects brought by the use of negative words being modified by hedge words. But in terms of positive words being modified by hedge words, it is a strategy of counterbalance to 'the excessive use of positive words and thus invite [s] the readers, including peers, politicians, journalists and citizens, to be cautious' (Bordignon et al., 2021, p. 12). What makes the situation more complex is the additional factor of finer disciplinary classification into 27 high and low ranking categories. According to Hyland (1998b), the so-called hard-soft distinction should be conceived as a continuum, which might be more convenient to explain the intersection of soft-science and hard-science journals in terms of both ranking and choice of words. Discourse communities, a concept developed by Swales (1990), referring to groups that have 'a broadly agreed set of common goals' (p. 26) help to explain the situation by stating that 'disciplinary communities construct their own conventions and modes of interaction through the writing they produce' (McGrath & Kuteeva, 2012, p. 171). That is to say, different high or low ranking categories form different discourse communities characterized by their own 'established conventions and understandings shared by community members' to guide their practices (McGrath & Kuteeva, 2012, p. 171). One of these practices is the linguistic and 'organizational patterns of written communication', also known as the genre that both defines and is defined by a particular community (Borg, 2003, p. 398).

Disciplines and rankings

First, as to the cross-disciplinary results, the present study corroborates Martín and Pérez (2014) observation that hard sciences tend to use more promotional discourse. According to Monteiro and Hirano (2020), competition and market value are two factors that might motivate scholars in hard sciences to promote their works more than their soft-science counterparts. That is to say, more competitive challenges in hard sciences increase the publication frequency in contrast to the 'reduced productivity' in soft sciences (Monteiro & Hirano, 2020, p. 17), and 'market value assigned to multiple fields in the hard sciences also strengthens the publishing disparities between hard and soft sciences' (Monteiro & Hirano, 2020, p. 17). A paradoxical result was provided by Fanelli (2010) who found that softer sciences in the hierarchy of sciences, with physical sciences at the top, social sciences at the bottom, and biological sciences in-between, report more positive outcomes, although both soft and hard sciences tend to report more positive results (over 80%) comparing with negative results. The discrepancy between the publication bias and linguistic positivity bias could be attributed to the speculation that hard sciences tend to linguistically encode the positive results more frequently than soft sciences. The present study also extends the previous study on explicit evaluation used in abstracts between narrative and hard sciences stating that although explicit evaluation in the humanities abstracts did not differ significantly from those in the social and natural sciences, evaluative attributes, the most common group of explicit evaluation containing lexis such as adjectives and nouns, 'were twice as common in the humanities and social science abstracts as in those from the natural sciences' (Stotesbury, 2003, p. 332). To

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put it in another way, positive words are prevailing in hard-science abstracts over those in soft sciences and negative words in both disciplines although smaller in number in contrast to positive words, when both positive and negative words are adjectives as explicit evaluation in the present study. However, the result of positive words superficially runs counter to the result of boosters from Hyland (1998b) that the number of boosters occurred less in the science and engineering papers than in humanities/social science papers, 'although the overall discrepancies were generally not so large' (Hyland, 1998b, p. 357). The main reason for this disparity lies in the method, particularly the list of boosters, adopted in Hyland (1998b), which was selected based on pragmatic and functional criteria in contrast to hedges, while the present study selected the list of positive words based mainly on semantic criteria in contrast to negative words and both lists of positive words and negative words were considered in contrast to hedges. In other words, the booster in Hyland (1998b) is a different concept from the positive word in the present study with the former being more like hedge in the present study. The use of hedge in the present study will be discussed below.

Second, more frequency of positive words in high ranking journals is possibly associated with citation bias (Gøtzsche, 2022), which is first nested within the context of the publication bias and outcome report bias (Dwan et al., 2013) and then lead to linguistic positivity bias in academic publications (Wen & Lei, 2022). The above-mentioned publication bias and outcome report bias seem to be associated with journal impact factors to a certain degree. Papers with positive results are 'currently published in journals with a higher impact factor' in some fields such as ophthalmology (Mimouni et al., 2015, p. 858), anaesthesiology (De Oliveira et al., 2012), oncology (Tang et al., 2014) and ecology (Ceausu et al., 2018), although the publication of positive or negative results might show no differences in term of journal impact factors in other fields such as clinical trials (Suñé et al., 2013), climate change (Harlos et al., 2017), paediatric ophthalmology (Gershoni et al., 2018), and retina randomized controlled trials (Parikh et al., 2021). Furthermore, positive results, especially those in journals with high impact factors (Etter & Stapleton, 2009), generally receive more citations than negative results (Duyx et al., 2017; Jannot et al., 2013; Koletsi et al., 2009; Nieminen et al., 2007). Given the citation bias nested within the context of publication bias positive words encoding the positive results might be more frequently used as shown in the results that a larger probability of using more positive words occurs in the high ranking journals with a significant difference but small effect size. In this way, it seems to create a 'virtuous circle' that high ranking journals as measured by journals' high citations and JIFs (Garfield, 1999) prefer positive reporting that may have more positive wording, in turn more positive reporting containing more positive wording may lead to higher citations (Duyx et al., 2017, p. 99), and eventually articles with positive wording are more likely to be accepted in such journals, which might be also reflected in their abstracts. Just as Evangelou et al. (2012) have speculated that journals with high impact factors usually prefer positive results, that is, the so-called 'selection bias may be stronger for high-IF journals' (p. 1274). The 'virtuous circle' might have a psychological effect on authors, reviewers, and editors 'to modulate the content of journals, particularly the preferential publication of studies describing significant findings' (Koletsi et al., 2009, p. 632.e3). Authors endeavour to publish significant results in high ranking journals to improve their credentials, and editors and reviews prefer 'an interesting topic and a favorable result' (Koletsi et al., 2009, p. 632.e4). To put it in another way, journals with high impact factors tend to have low acceptance rates (Sugimoto et al., 2013), 'which, in turn, raises the profile of the journal and enhances its likelihood of being cited' (Sugimoto et al., 2013, p. 904). As a result, more positive results are encoded linguistically to increase the chance of publication and citation, especially in high ranking journals due to the high pressure of competition and publication.

Third, the result of more hedges in soft-science abstracts corroborates that of Hyland (1998b), especially in philosophy and marketing (Hyland, 1998b). We also found more hedge words in philosophy and economics despite the different lists of hedge words being used. This disciplinary propensity has to do with different paradigms of the social construction of knowledge that hedges are used more frequently in soft sciences due to the fact that 'soft-knowledge areas however are typically more interpretative and less abstract' characterized by heterogeneity at various contextual levels such as 'less control of variables', 'more diversity of research outcomes', and negotiations of arguments between writers and readers (Hyland, 1998b, p. 361). This argument could also explain our results that more hedge words were used in astronomy, a hard science field, because apart from large amount of 'unknown unknowns', researchers in astronomy may mainly deal with 'known unknowns', the sources of uncertainty in the analysis including statistical, systematic, numerical, and so on (Hatfield, 2022). These uncertainties might result in more use of hedge words in astronomy. The result of more hedges in abstracts from high ranking journals is supported by Pei and Jurgens's (2021) view that 'higher-impact journals are described with comparatively less certainty' (p. 9966) partly encoded by hedges. But this result disagrees with those of Gross and Chesley (2012) and Bongelli et al. (2019). The possible reasons for this discrepancy are that their corpora consisted of medical articles in contrast to the abstracts from both soft and hard science journals in the present study and that the sizes of their corpora were relatively small. More hedge words related to high impact journals may be explained by stricter reviewing processes (Pei & Jurgens, 2021; Vinkers et al., 2015) and a common strategy of modifying positive words to show caution and of modifying negative words to show politeness.

Implications and limitations

As to the implications of the present study, first, it calls for more cautious use of language for writers and a wiser judgement of a paper from both reviewers and readers. Overusing positive words to promote results might lead to academic hype rendering less objective reports, which will, in turn, mislead readers and

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practitioners to take inappropriate actions. For the journals and reviewers, judging a report by its 'positive' results may lead to a distorted impression of a study (Boutron & Ravaud, 2018). It is worth noting that hedge words are not free from innocence, and maybe even worse, lead to 'scientific deceit' (John, 2021). Second, the linguistic encoding of scientific reports might indirectly influence the publication practices, to be specific, the disappearing of negative results (Fanelli, 2012). But negative results also matter (Gupta & Stopfer, 2011) for two reasons. First, the publication of negative results would 'help to save resources and foster scientific communication' (Gumpenberger et al., 2013, p. 277). Second, the value of scientific information is uncertain 'both immediately and in the long term', be it positive or negative results, because '[u]sually we do not know what information will be most useful (valuable) eventually' (Young et al., 2008, p. 1421).

Four limitations are noted here in the hope of being overcome in future studies. First, the present study is a semantically based analysis of frequency patterns. Further studies might conduct pragmatically based analysis of attitudinal stances of abstracts, especially sentiment analysis of abstracts based on a larger dataset. Second, another related factor that should be considered is that future studies will describe the modification patterns of positive and negative words by hedge words in greater detail and the collocation patterns of positive and negative words as well. Third, considering the complexity of interactive influence from various factors, the description of this situation is still relatively coarse, especially the influences from disciplinary knowledge and the 'virtuous circle' of journal ranking's influence on the word frequencies or vice versa. In the future, studies might focus on a particular discipline to have a detailed description and put journal ranking in a broader context as an independent topic of its own. Moreover, the present study takes the JIFs and JCIs of 2019 as a reference based on the premise of generally slow and weak variances from 2017 to 2021. However, it is also possible that JIFs or JCIs of very few journals change greatly. As to this issue, it is suggested that an appropriate impact factor should be selected as a reference according to the research questions. Fourth, although the so-called Pollyanna hypothesis has already been testified in both non-academic and academic fields as a seemingly universal feature of human language, this hypothesis is far from being determined. Therefore, it needs further investigation in the academic field cross-culturally by observing positive and negative words in academic publications from other languages.

AUTHOR CONTRIBUTIONS

Shaoliang Xie: Conceptualization; Methodology; Data Collection, Processing, and Analysis; Writing Original Draft; Writing Review & Editing; Software. **Chenggang Mi:** Software.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article:

Appendix S1. Supporting information

REFERENCES

Abdollahzadeh, E. (2019). A cross-cultural study of hedging in discussion sections by junior and senior academic writers. *Ibérica*, *38*, 177–202.

Bal-Gezegin, B., & Baş, M. (2020). Metadiscourse in academic writing: A comparison of research articles and book reviews. *Eurasian Journal of Applied Linguistics*, *6*(1), 45–62. https://doi.org/10.32601/ejal.710204

Barrass, R. (2002). *Scientists must write: A guide to better writing for scientists, engineers and students* (2nd ed.). Routledge.

Becher, T. (1989). *Academic tribes and territories: Intellectual inquiry and the cultures of disciplines*. SRHE/OUP.

Becher, T. (1994). The significance of disciplinary differences. *Studies in Higher Education*, *19*(2), 151–161. https://doi.org/10.1080/03075079412331382007

Begg, C. B., & Berlin, J. A. (1988). Publication bias: A problem in interpreting medical data. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, *151*(3), 419–445. https://doi.org/10.2307/2982993

Biglan, A. (1973). The characteristics of subject matter in different academic areas. *Journal of Applied Psychology*, *57*(3), 195–203. https://doi.org/10.1037/h0034701

Birkle, C., Pendlebury, D. A., Schnell, J., & Adams, J. (2020). Web of science as a data source for research on scientific and scholarly activity. *Quantitative Science Studies*, *1*(1), 363–376. https://doi.org/10.1162/qss_a_00018

Bongelli, R., Riccioni, I., Burro, R., & Zuczkowski, A. (2019). Writers' uncertainty in scientific and popular biomedical articles. A comparative analysis of the British Medical Journal and discover magazine. *PLoS One*, *14*(9), e0221933. https://doi.org/10.1371/journal.pone.0221933

Bordignon, F., Ermakova, L., & Noel, M. (2021). Over-promotion and caution in abstracts of preprints during the COVID-19 crisis. *Learned Publishing*, *34*(4), 622–636. https://doi.org/10.1002/leap.1411

Borg, E. (2003). Discourse community. *ELT Journal*, *57*(4), 398–400. https://doi.org/10.1093/elt/57.4.398

Boucher, J., & Osgood, C. E. (1969). The Pollyanna hypothesis. *Journal of Verbal Learning and Verbal Behavior*, *8*(1), 1–8. https://doi.org/10.1016/S0022-5371(69)80002-2

Boutron, I., & Ravaud, P. (2018). Misrepresentation and distortion of research in biomedical literature. *Proceedings of the National Academy of Sciences*, *115*(11), 2613–2619. https://doi.org/10.1073/pnas.1710755115

Brezina, V., & Gablasova, D. (2015). Is there a core general vocabulary? Introducing the new general service list. *Applied Linguistics*, *36*(1), 1–22. https://doi.org/10.1093/applin/amt018

Browne, C., Culligan, B., & Phillips, J. (2013). New general service list. www.newgeneralservicelist.org/

Cao, X., Lei, L., & Wen, J. (2021). Promoting science with linguistic devices: A large-scale study of positive and negative words in

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17414857, 2023, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/leap.1515 by Cochrane Philippines, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

academic writing. *Learned Publishing*, *34*(2), 82–88. https://doi.org/10.1002/leap.1322

Caulfield, T., & Condit, C. (2012). Science and the sources of hype. *Public Health Genomics*, *15*(3–4), 209–217. https://doi.org/10.1159/000336533

Ceausu, S., Borda-de-Agua, L., Merckx, T., Sossai, E., Sapage, M., Miranda, M., & Pereira, H. M. (2018). High impact journals in ecology cover proportionally more statistically significant findings. *bioRxiv*, 311068. https://doi.org/10.1101/311068

Coxhead, A. (2000). A new academic word list. *TESOL Quarterly*, *34*(2), 213–238. https://doi.org/10.2307/3587951

Crompton, P. (1997). Hedging in academic writing: Some theoretical problems. *English for Specific Purposes*, *16*(4), 271–287. https://doi.org/10.1016/S0889-4906(97)00007-0

Dang, T. N. Y. (2018). The nature of vocabulary in academic speech of hard and soft-sciences. *English for Specific Purposes*, *51*, 69–83. https://doi.org/10.1016/j.esp.2018.03.004

De Oliveira, G. S., Chang, R., Kendall, M. C., Fitzgerald, P. C., & McCarthy, R. J. (2012). Publication bias in the anesthesiology literature. *Anesthesia & Analgesia*, *114*(5), 1042–1048. https://doi.org/10.1213/ANE.0b013e3182468fc6

Dodds, P. S., Clark, E. M., Desu, S., Frank, M. R., Reagan, A. J., Williams, J. R., Mitchell, L., Harris, K. D., Kloumann, I. M., Bagrow, J. P., Megerdoomian, K., McMahon, M. T., Tivnan, B. F., & Danforth, C. M. (2015). Human language reveals a universal positivity bias. *Proceedings of the National Academy of Sciences*, *112*(8), 2389–2394. https://doi.org/10.1073/pnas.1411678112

Dos Santos, M. B. (1996). The textual organization of research paper abstracts in applied linguistics. *Text*, *16*(4), 481–499. https://doi.org/10.1515/text.1.1996.16.4.481

Duyx, B., Urlings, M. J., Swaen, G. M., Bouter, L. M., & Zeegers, M. P. (2017). Scientific citations favor positive results: A systematic review and meta-analysis. *Journal of Clinical Epidemiology*, *88*, 92–101. https://doi.org/10.1016/j.jclinepi.2017.06.002

Dwan, K., Gamble, C., Williamson, P. R., & Kirkham, J. J. (2013). Systematic review of the empirical evidence of study publication bias and outcome reporting bias—An updated review. *PLoS One*, *8*(7), e66844. https://doi.org/10.1371/journal.pone.0066844

Easterbrook, P. J., Gopalan, R., Berlin, J. A., & Matthews, D. R. (1991). Publication bias in clinical research. *The Lancet*, *337*(8746), 867–872. https://doi.org/10.1016/0140-6736(91)90201-Y

Ebrahimi, S. F., & Chan, S. H. (2015). Research article abstracts in applied linguistics and economics: Functional analysis of the grammatical subject. *Australian Journal of Linguistics*, *35*(4), 381–397. https://doi.org/10.1080/07268602.2015.1070660

Etter, J. F., & Stapleton, J. (2009). Citations to trials of nicotine replacement therapy were biased toward positive results and high-impact-factor journals. *Journal of Clinical Epidemiology*, *62*(8), 831–837. https://doi.org/10.1016/j.jclinepi.2008.09.015

Evangelou, E., Siontis, K. C., Pfeiffer, T., & Ioannidis, J. P. (2012). Perceived information gain from randomized trials correlates with publication in high-impact factor journals. *Journal of Clinical Epidemiology*, *65*(12), 1274–1281. https://doi.org/10.1016/j.jclinepi.2012.06.009

Fanelli, D. (2010). "Positive" results increase down the hierarchy of the sciences. *PLoS One*, *5*(4), e10068. https://doi.org/10.1371/journal.pone.0010068

Fanelli, D. (2012). Negative results are disappearing from most disciplines and countries. *Scientometrics*, *90*(3), 891–904. https://doi.org/10.1007/s11192-011-0494-7

Franco, A., Malhotra, N., & Simonovits, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, *345*(6203), 1502–1505. www.science.org/doi/10.1126/science.1255484

Fraser, B. (1980). Conversational mitigation. *Journal of Pragmatics*, *4*(4), 341–350. https://doi.org/10.1016/0378-2166(80)90029-6

Fraser, V. J., & Martin, J. G. (2009). Marketing data: Has the rise of impact factor led to the fall of objective language in the scientific article? *Respiratory Research*, *10*(1), 1–5. https://doi.org/10.1186/1465-9921-10-35

Gao, X. (2016). A cross-disciplinary corpus-based study on English and Chinese native speakers' use of linking adverbials in academic writing. *Journal of English for Academic Purposes*, *24*, 14–28. https://doi.org/10.1016/j.jeap.2016.08.002

Garfield, E. (1999). Journal impact factor: A brief review. *CMAJ*, *161*(8), 979–980.

Gerber, A. S., & Malhotra, N. (2008). Publication bias in empirical sociological research: Do arbitrary significance levels distort published results? *Sociological Methods & Research*, *37*(1), 3–30. https://doi.org/10.1177/0049124108318973

Gershoni, A., Ishai, M. B., Vainer, I., Mimouni, M., & Mezer, E. (2018). Positive results bias in pediatric ophthalmology scientific publications. *Journal of American Association for Pediatric Ophthalmology and Strabismus*, *22*(5), 394–395. https://doi.org/10.1016/j.jaapos.2018.03.012

Gøtzsche, P. C. (2022). Citation bias: Questionable research practice or scientific misconduct? *Journal of the Royal Society of Medicine*, *115*(1), 31–35. https://doi.org/10.1177/01410768221075881

Gross, A. G., & Chesley, P. (2012). Hedging, stance and voice in medical research articles. In K. Hyland & C. S. Guinda (Eds.), *Stance and voice in written academic genres* (pp. 85–100). Palgrave. https://doi.org/10.1057/9781137030825_6

Gumpenberger, C., Gorraiz, J., Wieland, M., Roche, I., Schiebel, E., Besagni, D., & François, C. (2013). Exploring the bibliometric and semantic nature of negative results. *Scientometrics*, *95*(1), 277–297. https://doi.org/10.1007/s11192-012-0829-z

Gupta, N., & Stopfer, M. (2011). Negative results need airing too. *Nature*, *470*(7332), 39. https://doi.org/10.1038/470039a

Harlos, C., Edgell, T. C., & Hollander, J. (2017). No evidence of publication bias in climate change science. *Climatic Change*, *140*(3), 375–385. https://doi.org/10.1007/s10584-016-1880-1

Hartley, J. (2003). Improving the clarity of journal abstracts in psychology: The case for structure. *Science Communication*, *24*(3), 366–379. https://doi.org/10.1177/1075547002250301

Hatfield, P. (2022). Quantification of Unknown Unknowns in Astronomy and Physics. *arXiv preprint*, arXiv:2207.13993. https://doi.org/10.48550/arXiv.2207.13993

Hedges, L. V. (1987). How hard is hard science, how soft is soft science? The empirical cumulativeness of research. *American Psychologist*, *42*(5), 443–455. https://doi.org/10.1037/0003-066X.42.5.443

Hu, G., & Cao, F. (2011). Hedging and boosting in abstracts of applied linguistics articles: A comparative study of English-and Chinese-medium journals. *Journal of Pragmatics*, *43*(11), 2795–2809. https://doi.org/10.1016/j.pragma.2011.04.007

Hyland, K. (1994). Hedging in academic writing and EAF textbooks. *English for Specific Purposes*, *13*(3), 239–256. https://doi.org/10.1016/0889-4906(94)90004-3

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Hyland, K. (1998a). *Hedging in scientific research articles*. John Benjamins Publishing Company. https://doi.org/10.1075/pbns.54

Hyland, K. (1998b). Boosting, hedging and the negotiation of academic knowledge. *Text & Talk*, *18*(3), 349–382. https://doi.org/10.1515/text.1.1998.18.3.349

Hyland, K. (2002). Genre: Language, context, and literacy. *Annual Review of Applied Linguistics*, *22*(1), 113–135. https://doi.org/10.1017/S0267190502000065

Hyland, K., & Jiang, F. K. (2021). 'Our striking results demonstrate…': Persuasion and the growth of academic hype. *Journal of Pragmatics*, *182*, 189–202. https://doi.org/10.1016/j.pragma.2021.06.018

Jannot, A. S., Agoritsas, T., Gayet-Ageron, A., & Perneger, T. V. (2013). Citation bias favoring statistically significant studies was present in medical research. *Journal of Clinical Epidemiology*, *66*(3), 296–301. https://doi.org/10.1016/j.jclinepi.2012.09.015

Jiang, F. K., & Hyland, K. (2017). Metadiscursive nouns: Interaction and cohesion in abstract moves. *English for Specific Purposes*, *46*, 1–14. https://doi.org/10.1016/j.esp.2016.11.001

John, S. (2021). Scientific deceit. *Synthese*, *198*(1), 373–394. https://doi.org/10.1007/s11229-018-02017-4

Khany, R., & Malmir, B. (2020). A move-marker list: A study of rhetorical move-lexis linguistic realizations of research article abstracts in social and behavioural sciences. *RELC Journal*, *51*(3), 381–396. https://doi.org/10.1177/0033688219833131

Koletsi, D., Karagianni, A., Pandis, N., Makou, M., Polychronopoulou, A., & Eliades, T. (2009). Are studies reporting significant results more likely to be published? *American Journal of Orthodontics and Dentofacial Orthopedics*, *136*(5), 632.e1–632.e5. https://doi.org/10.1016/j.ajodo.2009.06.002

Litkowski, K. (2020). Preposition analysis using correspondence analysis. Working Paper 20-02. www.clres.com/online-papers/ca4pdep.pdf

Martín, P., & Pérez, I. K. L. (2014). Convincing peers of the value of one's research: A genre analysis of rhetorical promotion in academic texts. *English for Specific Purposes*, *34*, 1–13. https://doi.org/10.1016/j.esp.2013.09.002

McGrath, L., & Kuteeva, M. (2012). Stance and engagement in pure mathematics research articles: Linking discourse features to disciplinary practices. *English for Specific Purposes*, *31*(3), 161–173. https://doi.org/10.1016/j.esp.2011.11.002

Millar, N., Budgell, B., & Salager-Meyer, F. (2020). Hype in reports of clinical research: The authors' perspectives. *English for Specific Purposes*, *60*, 53–64. https://doi.org/10.1016/j.esp.2020.07.001

Millar, N., Salager-Meyer, F., & Budgell, B. (2019). "It is important to reinforce the importance of…": 'Hype' in reports of randomized controlled trials. *English for Specific Purposes*, *54*, 139–151. https://doi.org/10.1016/j.esp.2019.02.004

Miller, A. S. (1969). The myth of objectivity in legal research and writing. *Catholic University Law Review*, *18*(3), 290–307.

Mimouni, M., Krauthammer, M., Gershoni, A., Mimouni, F., & Nesher, R. (2015). Positive results bias and impact factor in ophthalmology. *Current Eye Research*, *40*(8), 858–861. https://doi.org/10.3109/02713683.2014.957777

Monteiro, K., & Hirano, E. (2020). A periphery inside a semi-periphery: The uneven participation of Brazilian scholars in the international community. *English for Specific Purposes*, *58*, 15–29. https://doi.org/10.1016/j.esp.2019.11.001

Moreno, A. I. (2021). Selling research in RA discussion sections through English and Spanish: An intercultural rhetoric approach. *English for Specific Purposes*, *63*, 1–17. https://doi.org/10.1016/j.esp.2021.02.002

Moyano, E. I. (2019). Knowledge construction in discussions of research articles in two disciplines in Spanish: The role of resources of appraisal. *Journal of Pragmatics*, *139*, 231–246. https://doi.org/10.1016/j.pragma.2018.09.011

Nieminen, P., Rucker, G., Miettunen, J., Carpenter, J., & Schumacher, M. (2007). Statistically significant papers in psychiatry were cited more often than others. *Journal of Clinical Epidemiology*, *60*(9), 939–946. https://doi.org/10.1016/j.jclinepi.2006.11.014

Noguera, C. P. (2012). Writing business research article abstracts: A genre approach. *Ibérica*, *24*, 211–232.

Parikh, A., Markle, J., Venincasa, M., Kuriyan, A. E., Gupta, M., & Sridhar, J. (2021). Positive results bias and impact factor in retina clinical trials 2016 to 2019. *Retina*, *41*(8), 1697–1700. https://doi.org/10.1097/IAE.0000000000003107

Peacock, M. (2006). A cross-disciplinary comparison of boosting in research articles. *Corpora*, *1*(1), 61–84. https://doi.org/10.3366/cor.2006.1.1.61

Pei, J., & Jurgens, D. (2021). Measuring sentence-level and aspect-level (un)certainty in science Communications. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 9959–10011). https://aclanthology.org/2021.emnlp-main.784

Pérez-Llantada, C. (2003). Social pragmatics in technical writing: A corpus-based analysis of thematic articles. *Ibérica*, *5*, 19–34.

Pho, P. D. (2008). Research article abstracts in applied linguistics and educational technology: A study of linguistic realizations of rhetorical structure and authorial stance. *Discourse Studies*, *10*(2), 231–250. https://doi.org/10.1177/1461445607087010

Pigott, T. D., Valentine, J. C., Polanin, J. R., Williams, R. T., & Canada, D. D. (2013). Outcome-reporting bias in education research. *Educational Researcher*, *42*(8), 424–432. https://doi.org/10.1177/1461445607087010

Ribeiro, F. N., Araújo, M., Gonçalves, P., Gonçalves, M. A., & Benevenuto, F. (2016). Sentibench-a benchmark comparison of state-of-the-practice sentiment analysis methods. *EPJ Data Science*, *5*(1), 1–29. https://doi.org/10.1140/epjds/s13688-016-0085-1

Salager-Meyer, F. (1992). A text-type and move-analysis study of verb tense and modality distribution in medical English abstracts. *English for Specific Purposes*, *11*(2), 93–113. https://doi.org/10.1016/S0889-4906(05)80002-X

Serrano-Guerrero, J., Olivas, J. A., Romero, F. P., & Herrera-Viedma, E. (2015). Sentiment analysis: A review and comparative analysis of web services. *Information Sciences*, *311*, 18–38. https://doi.org/10.1016/j.ins.2015.03.040

Shapin, S. (2022). Hard science, soft science: A political history of a disciplinary array. *History of Science*, *60*(3), 287–328. https://doi.org/10.1177/00732753221094739

Shinohara, K., Suganuma, A. M., Imai, H., Takeshima, N., Hayasaka, Y., & Furukawa, T. A. (2017). Overstatements in abstract conclusions claiming effectiveness of interventions in psychiatry: A meta-epidemiological investigation. *PLoS One*, *12*(9), e0184786. https://doi.org/10.1371/journal.pone.0184786

Storer, N. W. (1967). The hard sciences and the soft: Some sociological observations. *Bulletin of the Medical Library Association*, *55*(1), 75–84.

Stotesbury, H. (2003). Evaluation in research article abstracts in the narrative and hard sciences. *Journal of English for Academic Purposes*, *2*(4), 327–341. https://doi.org/10.1016/S1475-1585(03)00049-3

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Sugimoto, C. R., Lariviére, V., Ni, C., & Cronin, B. (2013). Journal acceptance rates: A cross-disciplinary analysis of variability and relationships with journal measures. *Journal of Informetrics*, *7*(4), 897–906. https://doi.org/10.1016/j.joi.2013.08.007

Suñé, P., Suñé, J. M., & Montoro, J. B. (2013). Positive outcomes influence the rate and time to publication, but not the impact factor of publications of clinical trial results. *PLoS One*, *8*(1), e54583. https://doi.org/10.1371/journal.pone.0054583

Swales, J. (1990). *Genre analysis: English in academic and research settings*. Cambridge University Press.

Szomszor, M. (2021). Introducing the Journal Citation Indicator: A new, field-normalized measurement of journal citation impact [Web log post]. https://clarivate.com/blog/introducing-the-journal-citation-indicator-a-new-field-normalized-measurement-of-journal-citation-impact/

Taboada, M., Brooke, J., Tofiloski, M., Voll, K., & Stede, M. (2011). Lexicon-based methods for sentiment analysis. *Computational Linguistics*, *37*(2), 267–307. https://doi.org/10.1162/COLI_a_00049

Tang, P. A., Pond, G. R., Welch, S., & Chen, E. X. (2014). Factors associated with publication of randomized phase III cancer trials in journals with a high impact factor. *Current Oncology*, *21*(4), e564–e572. https://doi.org/10.3747/co.21.1937

Tempest, D. (2005). The effect of journal title changes on impact factors. *Learned Publishing*, *18*(1), 57–62. https://doi.org/10.1087/0953151052801460

Vinkers, C. H., Tijdink, J. K., & Otte, W. M. (2015). Use of positive and negative words in scientific PubMed abstracts between 1974 and 2014: Retrospective analysis. *BMJ*, *351*, h6467. https://doi.org/10.1136/bmj.h6467

Wang, W., & Yang, C. (2015). Claiming centrality as promotion in applied linguistics research article introductions. *Journal of English for Academic Purposes*, *20*, 162–175. https://doi.org/10.1016/j.jeap.2015.05.002

Wen, J., & Lei, L. (2022). Linguistic positivity bias in academic writing: A large-scale diachronic study in life sciences across 50 years. *Applied Linguistics*, *43*(2), 340–364. https://doi.org/10.1093/applin/amab037

Williamson, P. R., Gamble, C., Altman, D. G., & Hutton, J. L. (2005). Outcome selection bias in meta-analysis. *Statistical Methods in Medical Research*, *14*(5), 515–524. https://doi.org/10.1191/0962280205sm415oa

Xie, J., Gong, K., Cheng, Y., & Ke, Q. (2019). The correlation between paper length and citations: A meta-analysis. *Scientometrics*, *118*(3), 763–786. https://doi.org/10.1007/s11192-019-03015-0

Yang, Y. (2013). Exploring linguistic and cultural variations in the use of hedges in English and Chinese scientific discourse. *Journal of Pragmatics*, *50*(1), 23–36. https://doi.org/10.1016/j.pragma.2013.01.008

Young, N. S., Ioannidis, J. P. A., & Al-Ubaydli, O. (2008). Why current publication practices may distort science. *PLoS Medicine*, *5*(10), e201. https://doi.org/10.1371/journal.pmed.0050201

Zobel, J. (2004). *Writing for computer science*. Springer. https://doi.org/10.1007/978-0-85729-422-7

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